H2S × GOOGLE CLOUD GEN AI HACKATHON 2026 · SUSTAINABILITY & SOCIAL IMPACT

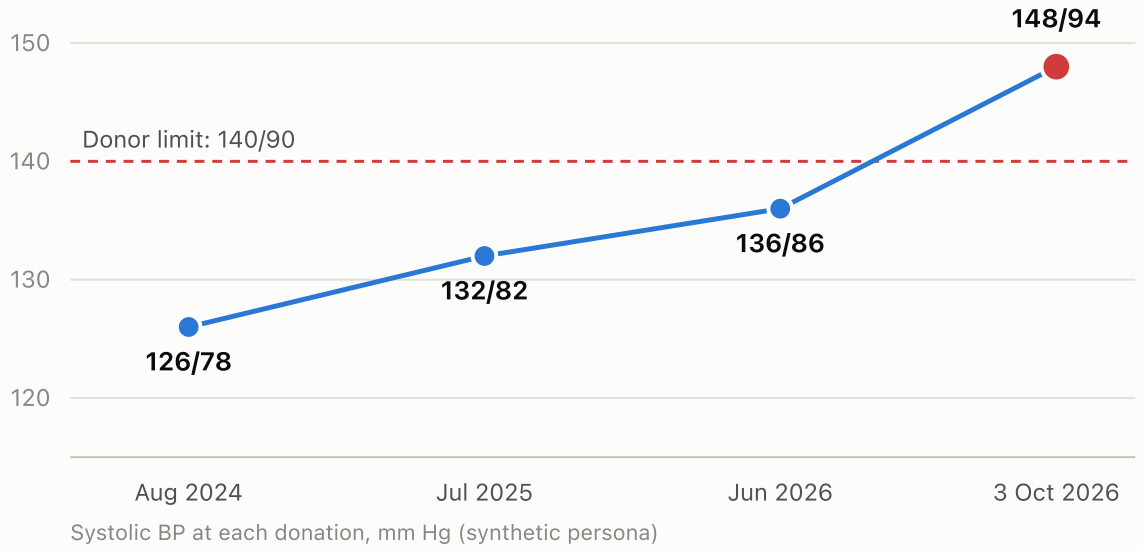
TraceDrop

Give blood. See every life it touches.

The donor's app: every part of your donation followed to the patient it helped, your free health check turned into a light follow-up until you can give again, and health-insurance cover that rises as you stay well.

THE MOMENT

He gave blood on 3 October. Five days later it helped a patient. He'll never know.



Systolic BP at each donation, mm Hg (synthetic persona)

Arjun, 34 · software engineer, Bengaluru

- Gives at office camps. His **platelets reached a patient** on 8 October. Nobody tells him.
- His BP has risen at **every donation**, each reading on a different form. Nobody joined the dots.
- 148/94 means he **can't give next time**. "Come back later." No reason he understands, no path back.

29%

of Indian adults aged 30–69 with raised BP **know they have it**.

ICMR-NCDIR National NCD Monitoring Survey

THE PROBLEM

Blood supply depends on donors who come back. Three things push them away.

India collects **1.57 crore** units a year. Repeat voluntary donors are the safest supply. Yet every donation ends in silence.

1 · The gift disappears

Up to three patients. Zero news.

A donation is split into red cells, platelets and plasma that reach different patients. The donor gets a certificate and silence.

Sweden texts donors when their blood is used, to bring them back. That text says only *that* it was used.

The Independent, 2015

2 · The health check goes nowhere

Measured every time. Explained never.

BP and haemoglobin are measured at every visit; the sample is screened. Donors who pass take nothing home.

Results alone don't help: Taiwan's donor screening found "limited benefits… without additional interventions".

Vox Sang 2025

3 · Turned away, then gone

"Come back later." They don't.

High BP or low haemoglobin means deferral, with no explanation and no path back. 57% of Indian women are anaemic.

Deferred first-time donors return at about half the rate of others.

NFHS-5; Custer 2007

Three reasons to give again, in one app, with one assistant: Drop.

Impact

See every life your blood touches

- Every part of every donation followed to the patient: the treatment and type of hospital.
- The patient stays anonymous: details shown only when at least 5 patients share them.
- A short **Story** when a part lands, ending on a shareable postcard.
- **Badges** for milestones; **Circles** for family, workplace and friends.

Health

Stay well until you can give again

- The donation's own results, in plain words.
- A finding becomes a **Goal**: one target, 2–3 steps, each citing its guidance.
- Check-ins every two weeks, check-up slots, reminders you ask for.
- The date you can donate again. At most one new Card a day.

Cover

Better cover for staying healthy

- Like a credit limit: cover steps **₹5 → 6 → 7 → 8 lakh** at the same premium.
- Lower risk from a better health score creates room for more cover.
- Only with your consent; the insurer sees the score, never results.
- Illustrative in the prototype.

Around the donor: what their blood meant for the blood centre, hospitals, labs and public health · a network-wide ecosystem view · guided booking with a Google Calendar event · **Your data**, which lists and deletes everything kept.

DEMO · LIVE ON CLOUD RUN

Arjun, from silence to a story, a Goal and a date to give again

10 Oct · Story

"My platelets reached a patient." Some details stay private so no one can be identified. He shares the postcard.

11 Oct · Results

BP up at each of his last 4 donations. One Goal: **BP under 130/80**. Walks, less salt, see a doctor within 3 weeks.

Ask Drop · Kannada

"Why can't I donate?" Drop looks it up and answers in two sentences, with his BP chart.

25 Oct · Check-in

He adds a pharmacy reading: 128/78. **Goal reached.**

Back to giving

"You can donate from 1 Jan." Drop will offer slots three weeks before.

Meera: turned away, then back

Hb 11.8, she needs 12.5. She photographs a lab report; **Gemini reads Hb 12.6**, she confirms, and she can donate again.

Vikram: a rule, not a model

BP 184/118. One urgent Card, always first, can't be hidden, only waits an hour: see a doctor today. His Goal suggests no exercise until a doctor has seen him.

HOW IT'S BUILT

Rules decide. Gemini helps where rules can't.

DONOR APP

React 19 web app
Home · Impact · Health · Cover · Community · Donate · Your data

Drop on every screen
Cards on Home, chat in a side sheet, the Intro

Google Sign-In · Calendar · Gmail
booking event and confirmation in the donor's own account

DATA

Synthetic world
300 donors · 657 visits · 1,627 components · seed 42

Firestore
accounts, sign-ins, preferences, bookings

GEMINI 3.1 FLASH-LITE · VERTEX AI

Drop's chat · agent with 18 tools
answers any question across impact, health and cover; shows a chart; offers a check-up; saves a reading; remembers a preference; six languages

Lab-report reader · multimodal
any lab's photo → 16 LOINC markers, unit conversion, schema + range checks, donor confirms

Intro translation
kept only if every number survives

Gateway
rate limit · daily budgets · traces · output re-validated · rule fallback for every feature

DETERMINISTIC ENGINES · 152 TESTS

Findings, score, eligibility
ACC/AHA 2017 and ICMR thresholds; NBTC gaps

Drop's Cards and Goals
Message budget, quiet hours, urgent first; targets and cited steps

Cover ladder
risk from score → headroom at the same premium

Recipient privacy
k = 5 generalisation, month-rounded dates

WHO WINS, AND HOW IT SCALES

The donor pays nothing. Everyone who needs donors to return has a reason to fund it.

Donors

- See what their blood did, privately
- Understand their own results
- A clear path back after a deferral
- Cover that rises as they stay well
- Control: Your data, Choices, delete

Institutions

- **Blood centres, SBTCs:** donors who return
- **Employers:** a wellness benefit on their group policy
- **Insurers:** lower-risk members, with consent
- **Public health:** an anonymised BP, sugar and anaemia signal
- **Patients:** a steadier blood supply

Path to scale

- Company camps at a few Bengaluru centres
- All donors at those centres
- SBTCs and eRaktKosh
- Other states and languages
- JAPAC blood services

Built to be trusted

No diagnosis or prescribing · urgent findings always say "see a doctor today" · patients anonymous by construction · insurer sharing off unless chosen · no inducement: deferred donors get the same follow-up · synthetic data only · every simulated part labelled as such.



Give blood.
See every life it touches.

When donors see what their gift did, stay healthy enough to give, and gain from staying well, they come back.
When they come back, patients have blood.